



# Vidyavardhini's College of Engineering & Technology

## Department of Civil Engineering

**Course Name:** Building Materials and Concrete Technology (2093113) **Sem/Year:** III/SE  
**Academic Year:** 2026-27

### Assignment No.: 03

Date of Submission:

Marks: 10

**2093113.3:** Students are able to select suitable concrete materials based on the properties of cement, lime, aggregates, and admixtures.

| Q.No | Questions                                                                                                                                                                                                                                                                        | BL |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Q.1) | A concrete mix is prepared using volume batching. Determine the bulking of the given sand sample and explain how the bulking phenomenon affects the quantity of sand used in concrete, with the help of a labelled sketch.                                                       | 3  |
| Q.2) | A concrete structure requires improved workability and durability. Select a suitable admixture for the given condition and explain how it improves the performance of concrete.                                                                                                  | 3  |
| Q.3) | A mass concrete foundation is being constructed where excessive heat generation may cause thermal cracking. Explain the effect of heat of hydration of Ordinary Portland Cement (OPC) and recommend appropriate measures by considering the relevant physical properties of OPC. | 3  |
| Q.4) | A heritage building requires lime plastering for restoration. Select the most suitable type of lime (fat lime, hydraulic lime, or quick lime) and explain its preparation method to achieve durability and breathability.                                                        | 3  |
| Q.5) | A high-strength concrete mix requires a low water-cement ratio without reducing workability. Select an appropriate chemical admixture and explain how it achieves the required concrete properties.                                                                              | 3  |
| Q.6) | A construction project in a coastal region plans to use manufactured sand (M-sand) instead of natural river sand. Assess the effect of the angular particle shape of M-sand on the workability of concrete and recommend suitable measures to maintain the desired workability.  | 3  |

Signature of Subject Incharge